Niti Mishra

AI Engineer Intern

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PORTFOLIO https://portfolio-jirihofman.vercel.app/?customUsername=nitimishra

STATEMENT

AI/ML-focused final year CS undergrad skilled in Python, PyTorch, LLMs, LangChain, and RAG. Experienced in building production-ready AI applications to solve real-world problems. Seeking an AI/ML opportunity to drive impact and expand technical expertise.

LINKS

LinkedIn
https://www.linkedin.com/in/niti-mishra21

GitHub
https://github.com/nitimishra

EDUCATION

APJ Abdul Kalam Technical University
In progress
Bachelor's Degree Computer Systems Engineering

Kendriya Vidyalaya IFFCO Phulpur
2020 — 2022
Intermediate (Class XII)

SKILLS TRIDENT

AI / ML
Machine Learning · Deep Learning · Neural Networks · Data Preprocessing · Feature Engineering · Model Evaluation

Languages
Python · SQL · C

Gen AI & LLMs
Large Language Models (LLMs) · LangChain · LangSmith · Retrieval-Augmented Generation (RAG) · Transformers · Embeddings · Vector Search · Prompt Engineering

Libraries
PyTorch · TensorFlow / Keras · OpenCV · NumPy · Pandas · Scikit-learn · Tkinter

MLOps & Deployment
MLflow · FastAPI · Streamlit · Docker · Model Deployment · Experiment Tracking

Data & Storage
Git · GitHub · Docker · VS Code · Jupyter Notebook

Professional
Problem Solving · Communication · Leadership · Time Management · Teamwork

LANGUAGES

English
Fluent

Hindi
Native

Work Experience

AI Research Intern • IIT Indore

July 2026 - August 2026 (2 Months) Indore

- Image Segmentation: Worked on image segmentation tasks to identify and classify specific regions or objects within images, gaining practical experience with computer vision and deep learning techniques.
- Model Training: Learned and applied the end-to-end process of training deep learning models, including data preparation, model configuration, training, validation, and performance evaluation.
- Data Annotation: Performed image data annotation and labeling to create high-quality datasets for supervised machine learning and computer vision model development.

Cloud Computing Virtual Intern • IBM

June 2026 – August 2026 Remote

- Completed a Virtual Internship in Cloud Computing through FSP (IBMCEP), gaining practical exposure to AWS and IBM Cloud.
- Worked with Docker and Kubernetes for containerization and application orchestration, along with cloud-based deployment.
- Gained hands-on understanding of CI/CD pipelines and DevOps workflows for automated application build, testing, and deployment.

PROJECTS

aerial-unet • Python • TensorFlow • Keras • U-Net • NumPy • Scikit-learn • PIL • Matplotlib

July 2026 – August 2026 https://github.com/nitimishra/aerial-unet

- Developed a custom U-Net model for pixel-level semantic segmentation of aerial imagery.
- Preprocessed 512×512 images and encoded segmentation masks into 6 semantic classes.
- Trained, evaluated, and visualized model predictions using TensorFlow/Keras.

Secure Password Generator Telegram Bot • python-telegram-bot • Cryptographic Randomness

Oct 2024 – Dec 2024 github.com/nitimishra/Telegram-bot

- Developed an interactive Telegram bot that generates five cryptographically secure passwords per user request.
- Leveraged Python's secrets module to ensure randomness, making passwords resistant to brute-force attacks.
- Provided user-defined password length, delivering a privacy-focused, chat-native security tool.

RAG Pipeline — Document Q&A Application • Python • LLMs • Retrieval-Augmented Generation • Docker

July 2026 github.com/nitimishra/RAG-Application

- Built a modular RAG pipeline with separate ingestion, retrieval, and answer-generation stages for document-grounded Q&A.
- Implemented a history-aware retrieval component that reformulates follow-up queries using prior conversation context.
- Used vector embeddings and similarity search to retrieve relevant document chunks, reducing LLM hallucination.
- Containerized the application with Docker for consistent, portable deployment.

Poetic-Text-Generation • Python • TensorFlow/Keras • LSTM • NLP

June 2026 github.com/nitimishra/Poetic-Text-Generation

- Developed a character-level LSTM-based text generation model trained on Shakespeare's corpus to produce stylistically coherent poetic text.
- Implemented temperature-based sampling and a custom preprocessing pipeline (character-to-index mapping, sequence windowing) to train a multi-layer LSTM capturing long-range dependencies in literary text.

CERTIFICATION

AI-Fluency: Framework & Foundations • Anthropic

August 2026 Remote



Claude 101 • Anthropic

August 2026 Remote



Bootcamp on Unmanned Aerial Systems • IIT Dhanbad

1 Month Prayagraj



Shortlisted Team (College Level) • Smart India Hackathon

2026 Prayagraj